Eric Yu

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Education

University of California, Berkeley May 2027
Bachelor's in Computer Science and Data Science *GPA: 3.74/4.0*

Relevant Coursework: Data Structures and Algorithms, Data Science, Linear Algebra, Discrete Math, Computer Architecture, Computer Security, Advanced Probability Theory, Deep Learning and Computer Vision, Quantitative Finance

Skills

Languages: Python, SQL, R, Java, JavaScript, C, C++, Ruby, x86, RISC-V, HTML, CSS
Data Science/Machine Learning: Pandas, NumPy, SciPy, Matplotlib, Scikit-Learn, PyTorch, TensorFlow, Hugging Face, OpenCV
Software Development: Git, React.js, Express.js, Tailwind CSS, Docker, Flask, AWS, MongoDB, PostgreSQL, OAuth, FastAPI

Experience

Incoming Software Engineer Intern 2026
PayPal - Compliance Platform Team *San Jose, CA*

Software Engineer Intern Jun 2025 - Aug 2025
Onward *Nashville, TN*

- Migrated the entire codebase from legacy styles to **Tailwind CSS v4** reducing full CSS build times by **5x** and incremental builds by **100x** significantly accelerating developer workflows
- Led a **PostgreSQL** production data migration involving 15+ million records, using batched **Sidekiq** jobs to ensure zero downtime and maintain data integrity
- Supported engineering team in improving internal tooling for Shopify integrations and merchant shop monitoring using **Sentry**
- Ensured feature correctness of 100 percent of deliverables by developing new integration tests in **RSpec** and expanding test automation, preventing future regressions

Undergraduate Researcher under Prof. Gopala Anumanchipalli Sep 2025 - Present
Berkeley Artificial Intelligence Research (BAIR) Lab - Berkeley Speech Group *Berkeley, CA*

- Leading the **Articulatory Gestures Project**, advancing gesture-based modeling for interpretable text-to-speech and robotic speech systems

Undergraduate Researcher under PhD Huiqi Wang Sep 2025 - Present
The Quantitative Ecosystem Dynamics Lab - Keenan Group *Berkeley, CA*

- Built an end-to-end geospatial retrieval pipeline by automating stratified sampling and preprocessing of multi-modal time-series data (imagery, terrain, climate, land cover, hazards), generating 3K+ aligned text–embedding training examples
- Trained and deployed a cosine-loss MLP to map 768-D text embeddings into Google AlphaEarth's 64-D satellite embedding space, enabling natural-language–driven location retrieval via cosine similarity and BFS-based contiguous region expansion

Machine Learning Engineer Intern April 2025 - Jun 2025
Anchor Logics (Skydeck accelerator-backed startup) *Berkeley, CA*

- Collaborated on designing and implementing an advanced gait analysis pipeline leveraging **OpenPose**, **DeepSORT**, and **OpenCV** to accurately extract key walking metrics such as stride length, cadence, and step height
- Built and deployed scalable **AWS** infrastructure (**EC2, S3**) to automate video processing and storage, and created a **React/FastAPI** web app for real-time video uploads and analysis

Autonomous Team Sep 2025 - Present
Formula Electric at Berkeley (Formula SAE) *Berkeley, CA*

- Developed an nonlinear model predictive control (MPC) framework using **CasADi** to optimize control inputs (acceleration, steering rate) for a nonlinear kinematic bicycle model under actuator and track boundary constraints
- Formulated and solved constrained trajectory optimization problems in real-time (30Hz) with IPOPT solver, ensuring stability and collision-free path tracking on a autonomous racing simulator

Projects

Real-time Graphics Engine August 2025

- Developed a 3D graphics engine in **C++** from scratch with a custom render loop, matrix-based transformations, and real-time scene updates
- Implemented OpenGL-based rendering pipeline with GLSL shaders, VBO/VAO management, Z-buffering, and texture sampling
- Built a scene graph architecture supporting hierarchical transformations, frustum culling, and spatial partitioning

RISC-V Neural Network June 2025

- Implemented a neural network using **RISC-V** assembly to classify handwritten digits, achieving 93% accuracy in the test set

Leadership

CS61A Academic Intern February 2025 - May 2025

- Assisted in teaching over 85 students in weekly labs programming concepts in OOP, Recursion, and Data Abstraction